Project: Worked examples
Title: Enclosure heat dissipation and fan sizing
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Total heat load inside a control enclosure, the effective surface that can dissipate it for the mounting condition, the internal temperature rise with natural convection and radiation through a stated overall heat-transfer coefficient, the forced-ventilation airflow for a target rise (density corrected for altitude), the fan duty point against a fan curve and the system resistance, optional solar gain, and the cooling-unit capacity needed when the limit cannot be met.
P = Σ P_i n_i d_i (+ solar α I A_sun). A_eff = Σ f_face × A_face. Natural: ΔT = P/(h A_eff) (a plain sum of exposed face areas times one coefficient; the IEC TR 60890 tables are not used). Forced: V = P/(ρ c_p ΔT_target) with ρ = p(z)/(287.055 T) and c_p = 1006 J/kg·K; fan duty where p_fan(V) = p_sys (V/V_ref)² (bisection). Cooling unit: P_cool = P − h A_eff (T_max − T_amb) when positive. Sealed enclosure (T_seal entered, no breather): the pressure difference across the walls when the air warms from the sealing temperature to T_int at constant volume, Δp = p_site·|T_int/T_seal − 1| (absolute temperatures), a rise when T_int > T_seal, a suction when below. A pressure-compensation element (breather or relief) limits it: dp_seal = min(that difference, dp_relief), the cracking pressure. The force of dp_seal on a cover of area A_cover is F_seal = dp_seal·A_cover.
Ref: IEC TR 60890 — A method of temperature-rise verification of low-voltage switchgear and controlgear assemblies by calculation
Ref: Manufacturer enclosure climate-control guides (Rittal, nVent Hoffman) — Effective surface, heat-transfer coefficient and fan-sizing practice (cited, not copied)
Ref: BS EN 61439-1 10.10 — Verification of temperature rise

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
loads	Heat loads	2 row(s)		spec: Panel schedule PNL-04	any
loads[1]		name = PLC and I/O; P = 80 W; qty = 1; duty = 1			
loads[2]		name = drives; P = 55 W; qty = 4; duty = 1			
P_add	Additional heat load (linked from E6's dissipated heat or another calculation)	0	W	default	0 … 100000 W
I_sun	Solar irradiance on the enclosure (0 = indoors)	0	W/m^2	default	0 … 1200 W/m^2
A_sun	Sunlit area	0	m^2	default	0 … 100 m^2
alpha_sun	Surface absorptivity	0.5		default	0.1 … 1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
W	Enclosure width	0.6	m	spec: Enclosure datasheet AE-1380: 600 × 800 × 300	0.1 … 5 m
H	Enclosure height	0.8	m	spec: Enclosure datasheet AE-1380: 600 × 800 × 300	0.1 … 5 m
D	Enclosure depth	0.3	m	spec: Enclosure datasheet AE-1380: 600 × 800 × 300	0.1 … 5 m
f_top	Top exposure (1 free, 0 covered)	1		default	0 … 1
f_front	Front exposure	1		default	0 … 1
f_rear	Rear exposure (0 = wall-mounted)	0		default	0 … 1
f_left	Left side exposure	1		default	0 … 1
f_right	Right side exposure	1		default	0 … 1
f_bottom	Bottom exposure (0 = on the floor)	0		default	0 … 1
h	Overall heat-transfer coefficient of the enclosure surface (editable reference value)	5.5	W/(m^2*K)	default	3 … 8 W/(m^2*K)
T_amb	Ambient temperature	35	degC	spec: Site conditions SC-2: 35 °C plant room; 50 °C inside limit for the drives	-20 … 55 degC
T_max	Maximum internal temperature	50	degC	spec: Site conditions SC-2: 35 °C plant room; 50 °C inside limit for the drives	20 … 80 degC
altitude	Site altitude (air density)	0	m	assumption: Wall mounted on a floor plinth: rear and bottom covered; indoors.	0 … 4000 m
V_pts	Fan curve: airflow points	[0, 100, 200]	m^3/h	spec: Fan datasheet FF-120: curve points	any
p_pts	Fan curve: static pressure points	[60, 30, 0]	Pa	spec: Fan datasheet FF-120: curve points	any
p_sys	System resistance (filters, grilles) at the reference airflow	20	Pa	spec: Filter datasheet: 20 Pa at 100 m³/h	0 … 500 Pa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
V_ref	Reference airflow for the system resistance	100	m^3/h	spec: Filter datasheet: 20 Pa at 100 m³/h	1 … 100000 m^3/h

3. Assumptions

A1. Heat leaves the enclosure through its exposed faces at one overall coefficient h (natural convection plus radiation to surroundings at the ambient temperature); covered faces dissipate nothing. (IEC TR 60890-style effective-surface approach; h is a reference value the user may edit.)
A2. Forced ventilation removes the full heat load with air entering at ambient and leaving at the target rise; natural dissipation is neglected for the airflow (conservative). (Manufacturer fan-sizing practice.)
A3. A sealed enclosure (T_seal entered) is a closed, rigid, leak-free volume of dry air at the site pressure when closed: the pressure difference follows from the ideal-gas relation at constant volume, from the sealing temperature to the internal temperature with natural dissipation T_int (the temperature without a cooling unit); a breather, a seal that leaks or a flexing wall reduces it, so it is an upper bound (a pressure-compensation element is modelled only as a cap at its cracking pressure, with no flow resistance); the site pressure is the standard atmosphere at the altitude. (Ideal gas at constant volume (derived).)
A4. The fan curve is linear between the entered points; the system resistance scales with airflow squared. (Duty-point idealisation.)
A5. altitude (Site altitude (air density)) is taken as 0 m. (Wall mounted on a floor plinth: rear and bottom covered; indoors.)

4. Working

[18] P_loads = ∑ P_i × n_i × d_i
= 2 item(s)
= 300 W
Internal heat load

[19] P_solar = alpha_sun * I_sun * A_sun
= 0.5 × 0 W/m² × 0 m²
= 0 W
Solar gain

[20] P_total = P_loads + P_add + P_solar
= 300 W + 0 W + 0 W
= 300 W
Total heat to remove

[21] A_eff = f_top * W * D + f_bottom * W * D + f_front * W * H + f_rear * W * H + f_left * H * D
IEC TR 60890 approach
= 1 × 0.6 m × 0.3 m + 0 × 0.6 m × 0.3 m + 1 × 0.6 m × 0.8 m + 0 × 0.6 m × 0.8 m + 1 × 0.8 m × 0.8 m × 0.3 m
= 1.14 m²
Effective dissipating surface

[22] dT_nat = P_total / (h * A_eff)
= 300 W / (5.5 W/(m²·K) × 1.14 m²)
= 47.847 K
Temperature rise with natural dissipation

[23] T_int = T_amb + dT_nat
= 35 degC + 47.847 K
= 82.847 degC
Internal temperature with natural dissipation

[24] dT_allow = T_max - T_amb
= 50 degC − 35 degC
= 15 K

Allowed rise

```
[25] u_nat = dT_nat / dT_allow
= 47.847 K / 15 K
= 3.1898
```

Natural dissipation utilisation

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[26] P_nat_max = h * A_eff * dT_allow
= 5.5 W/(m²·K) × 1.14 m² × 15 K
= 94.05 W
```

Heat the enclosure dissipates naturally at the limit

```
[27] P_cool = max(0 W, P_total - P_nat_max)
= max(0 W, 300 W - 94.05 W)
= 205.95 W
```

Cooling-unit capacity needed if ventilation is not used

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[28] p_site = 101325 (1 - 2.25577e-5 z)^5.2559
= z = 0 m
= 101.32 kPa
```

Site pressure

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[29] rho_air = p_site / (287.055 J/(kg*K) * T_amb)
= 101.32 kPa / (287.055 J/(kg*K) × 35 degC)
= 1.1455 kg/m³
```

Air density at ambient and altitude

```
[30] V_req = P_total / (rho_air * 1006 J/(kg*K) * dT_allow)
= 300 W / (1.1455 kg/m³ × 1006 J/(kg*K) × 15 K)
= 62.481 m³/h
```

Airflow for forced ventilation at the allowed rise (natural dissipation neglected)

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[31] V_req_net = max(0 m^3/h, (P_total - P_nat_max) / (rho_air * 1006 J/(kg*K) * dT_allow))
= max(0 m^3/h, (300 W - 94.05 W) / (1.1455 kg/m³ × 1006 J/(kg*K) × 15 K))
= 42.893 m³/h
```

Airflow crediting natural dissipation

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[36] V_duty = p_fan(V) = p_sys (V/V_ref)², bisection
=
= 113.75 m³/h
```

Fan duty airflow

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[37] p_duty = p_sys (V_duty/V_ref)²
= 20 × (113.7/100)²
= 25.876 Pa
```

Static pressure at duty

```
[38] u_fan = V_req / V_duty
= 62.481 m³/h / 113.75 m³/h
= 0.5493
```

Airflow utilisation (required / delivered)

5. Results

SYMBOL	VALUE	DESCRIPTION
P_loads	300 W	Internal heat load
P_solar	0 W	Solar gain
P_total	300 W	Total heat
A_eff	1.14 m²	Effective surface
dT_nat	47.847 K	Natural temperature rise
T_int	82.847 degC	Internal temperature (natural)
P_nat_max	94.05 W	Natural dissipation at the limit

SYMBOL	VALUE	DESCRIPTION
P_cool	205.95 W	Cooling unit needed
T_max	50 degC	Internal temperature limit
rho_air	1.1455 kg/m³	Air density
V_req	62.481 m³/h	Required airflow
V_req_net	42.893 m³/h	Required airflow (net)
V_duty	113.75 m³/h	Fan duty airflow
p_duty	25.876 Pa	Fan duty pressure
u_fan	0.5493	Airflow utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Natural dissipation holds T_max	u_nat = 3.1898	u_nat ≤ 1	3.19	
PASS	Fan delivers the required airflow	u_fan = 0.5493	u_fan ≤ 1	0.549	

7. Applicability

WARNING — input.P_add: P_add (Additional heat load (linked from E6's dissipated heat or another calculation)) was not entered; the template default 0 W is used.
WARNING — input.I_sun: I_sun (Solar irradiance on the enclosure (0 = indoors)) was not entered; the template default 0 W/m^2 is used.
WARNING — input.A_sun: A_sun (Sunlit area) was not entered; the template default 0 m^2 is used.
WARNING — input.alpha_sun: alpha_sun (Surface absorptivity) was not entered; the template default 0.5 is used.
WARNING — input.f_top: f_top (Top exposure (1 free, 0 covered)) was not entered; the template default 1 is used.
WARNING — input.f_front: f_front (Front exposure) was not entered; the template default 1 is used.
WARNING — input.f_rear: f_rear (Rear exposure (0 = wall-mounted)) was not entered; the template default 0 is used.
WARNING — input.f_left: f_left (Left side exposure) was not entered; the template default 1 is used.
WARNING — input.f_right: f_right (Right side exposure) was not entered; the template default 1 is used.
WARNING — input.f_bottom: f_bottom (Bottom exposure (0 = on the floor)) was not entered; the template default 0 is used.
WARNING — input.h: h (Overall heat-transfer coefficient of the enclosure surface (editable reference value)) was not entered; the template default 5.5 W/(m^2*K) is used.
WARNING — cooling-unit: T_max: natural dissipation holds 94.05 W of 300 W; ventilate or fit a 206 W cooling unit.

8. Verification

Template elec.enclosure-heat@1.4.0 Content hash 3add357ba3b44d85 Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0